

Non-Standard CSA Curves

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Abstract

In this document we summarize how to apply the FX Forward invariance to solve for non-standard CSA yield curves using the Mizuho MLIB analytics library. We illustrate this using GBP curves with a EUR CSA collateral as an example.

FX Forward Invariance

The Ratio of discount factors is invariant for any given CSA collateral currency and can be expressed as

$$\underbrace{\frac{P(t,T)^{GBP_USDCSA}}{P(t,T)^{EUR_USDCSA}}}_{\text{From Xccy Curves}} = \frac{P(t,T)^{GBP_EURCSA}}{P(t,T)^{EUR_EURCSA}} \quad (1)$$

CSA Curves

Rearranging (1) we can solve for any CSA curve using Xccy curves collateralized in USD and a Native OIS or Standard CSA curve

$$P(t,T)^{GBP_EURCSA} = \underbrace{\left(\frac{P(t,T)^{GBP_USDCSA}}{P(t,T)^{EUR_USDCSA}} \right)}_{\text{From Xccy Curves}} \times \underbrace{P(t,T)^{EUR_EURCSA}}_{\text{Standard CSA}} \quad (2)$$

MLIB CSA Curve Solver

In MLIB we represent (1) as

$$\frac{Leg1Forecast}{Leg2Forecast} = \frac{Leg1Discount}{Leg2Discount} \quad (3)$$

and solve for (2)

$$Leg1Discount = \frac{Leg1Forecast}{Leg2Forecast} \times Leg2Discount \quad (4)$$

Note that inverting leg 1 and leg 2 in (3) and (4) are mathematically equivalent.

$$\frac{Leg2Forecast}{Leg1Forecast} = \frac{Leg2Discount}{Leg1Discount} \quad (5)$$

and solve for (2)

$$Leg2Discount = \frac{Leg2Forecast}{Leg1Forecast} \times Leg1Discount \quad (6)$$